

# V3 empirical study · Instruction

One model · four underlyings · four volatility regimes · 1.5-year monthly calibration

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## What this report is

- Computational source of truth: the companion Jupyter notebook. This PDF is written from the same payload; every table number is identical.
- Question: which of 1 American-call pricing models tracks listed market prices most closely, and does the ranking change across companies and volatility regimes?
- In every table the BEST row is the lowest percentage RMSE. That row is bold and shaded green, and the Mark column says BEST.

## Experimental design (held fixed for every model)

- Models: GARCH(1,1).
- Companies: SPY (primary), AAPL, MSFT, AMZN. Each name is calibrated and priced separately.
- Regimes (V3 one-year evaluation windows): Crisis 2008-08-01→2009-07-31; Normal 2014-01-01→2014-12-31; Late-cycle 2018-10-01→2019-09-30; COVID 2019-09-01→2020-08-31.
- The late-cycle and COVID files share September 2019 (V3 notebook dates). All other regime days are disjoint. Contracts are built independently in each window.
- If history is shorter than the requested lookback, the estimator uses all available observations up to the update date (equity starts 2003-12-01; listed-option panels start 2008-01).

# V3 empirical study · Instruction (continued)

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## Shared market sample and filters (not model-specific)

- For each company  $\times$  regime, one listed-call sample is drawn once and reused by all one model in this report: nearest-ATM call on each Monday (next session if Monday is closed), DTE 7–60, listed expiry, as-of that date.
- Estimation filters, applied in order to every quote used for Heston/Heston–Merton calibration and to the LSM comparison set: (1) calls only, finite  $S$ ,  $K$ ,  $C$ ; (2) no-arbitrage  $C \geq \max(0, S-K)$ ; (3)  $7 \leq \text{DTE} \leq 60$ ; (4)  $|S/K - 1| \leq 10\%$ ; (5) premium  $\geq 0.05$ , valid bid-ask with relative spread  $\leq 50\%$  when those columns exist, volume  $\geq 1$  when present.
- Return-based models (GBM, Merton, GARCH, GARCH–Merton) never see option quotes in §4; they still price the same filtered LSM contracts in evaluation.

## Model-correctness adjustments (V3)

- Heston / Heston–Merton:  $(\kappa, \theta, \xi, \rho, v_0)$  are option-implied Fourier NLS (Albrecher little-trap), not Method A realized-variance moments.  $\mu$  is the P-measure lookback mean of stock returns; LSM replaces  $\mu \rightarrow r$ .
- Euler ordering: the stock increment over  $[t, t+\Delta t]$  uses the current variance  $v_t$ ;  $v_{t+\Delta t}$  is updated afterwards. No look-ahead in the variance.
- LSM: Longstaff–Schwartz American call on the full path cloud ( $n_{\text{paths}} = 10000$ , seed 42,  $n_{\text{steps}} = \text{DTE}$ ,  $\Delta t = 1/252$ ). Paths are not averaged before stopping.

## Metrics

- Primary:  $\text{RMSE\%} = 100 \times \sqrt{\text{mean}(((C_{\text{model}} - C_{\text{mkt}}) / C_{\text{mkt}})^2)}$ . Ranking uses this number only.
- Secondary:  $\text{MAE} = \text{mean} |C_{\text{model}} - C_{\text{mkt}}|$  (dollars).  $\text{Bias \$} = \text{mean}(C_{\text{model}} - C_{\text{mkt}})$  (dollars).  $\text{Bias\%} = 100 \times \text{mean}((C_{\text{model}} - C_{\text{mkt}})/C_{\text{mkt}})$ . Positive bias = model expensive vs market. Early-exercise fraction = mean share of LSM paths that stop before expiry.

## Page 3 · Filters, sample, and no-look-ahead

### Shared evaluation sample · n listed calls (identical for all one model in this report)

| Regime               | Window                  | SPY n | AAPL n | MSFT n | AMZN n |
|----------------------|-------------------------|-------|--------|--------|--------|
| 2008-2009 Crisis     | 2008-08-01 → 2009-07-31 | 50    | 50     | 50     | 50     |
| 2013-2014 Normal     | 2014-01-01 → 2014-12-31 | 50    | 50     | 50     | 50     |
| 2018-2019 Late-cycle | 2018-10-01 → 2019-09-30 | 50    | 51     | 50     | 51     |
| 2019-2020 COVID      | 2019-09-01 → 2020-08-31 | 50    | 50     | 50     | 50     |

### Filter funnel · last-step n in the 1.5-year lookback ending at each regime (processed call panel)

| Ticker | Regime    | Input* | No-arb | DTE 7-60 | S/K-1 ≤10% | Liquidity |
|--------|-----------|--------|--------|----------|------------|-----------|
| SPY    | 2008-2009 | 3359   | 3311   | 3311     | 3203       | 3203      |
| SPY    | 2013-2014 | 12815  | 12548  | 12548    | 12405      | 12405     |
| SPY    | 2018-2019 | 35222  | 35104  | 35104    | 34809      | 34809     |
| SPY    | 2019-2020 | 42100  | 41961  | 41961    | 41379      | 41379     |
| AAPL   | 2008-2009 | 685    | 685    | 685      | 652        | 652       |
| AAPL   | 2013-2014 | 8925   | 8875   | 8875     | 8633       | 8633      |
| AAPL   | 2018-2019 | 6309   | 6297   | 6297     | 6084       | 6084      |
| AAPL   | 2019-2020 | 8010   | 7992   | 7992     | 7713       | 7713      |
| MSFT   | 2008-2009 | 566    | 566    | 566      | 538        | 538       |
| MSFT   | 2013-2014 | 3299   | 3237   | 3237     | 3175       | 3175      |
| MSFT   | 2018-2019 | 7814   | 7752   | 7752     | 7583       | 7583      |
| MSFT   | 2019-2020 | 6892   | 6852   | 6852     | 6655       | 6655      |
| AMZN   | 2008-2009 | 465    | 465    | 465      | 442        | 442       |
| AMZN   | 2013-2014 | 5715   | 5692   | 5692     | 5598       | 5598      |
| AMZN   | 2018-2019 | 28809  | 28808  | 28808    | 28346      | 28346     |
| AMZN   | 2019-2020 | 26766  | 26760  | 26760    | 26323      | 26323     |

\*Input is the processed call panel restricted to (regime end – 1.5-year, regime end]. Processed files already drop many raw quotes; the funnel above is the additional V3 research filter.

No look-ahead. Monthly parameters at update date  $t$  use only equity returns and listed quotes with trading\_date  $\leq t$ . LSM prices the Monday contract using the latest calibration row with date  $\leq$  that Monday. Quotes are as-of the grid timestamp (same session, else the prior session).

SPY 2008–2009 uses bid-ask mid when the source mark is \$0.01 (a GitHub field error, not missing trades). After that repair the processed panel has weekly coverage, so  $n$  matches AAPL/MSFT (~50). All one model in this report still share that exact sample.

Heston / Heston-Merton calibration further subsamples the filtered lookback surface to  $\leq 24$  contracts (maturity  $\times$  moneyness strata). GBM / Merton / GARCH / GARCH-Merton estimate from lookback log returns only (jumps:  $3\sigma$  residual threshold where applicable).

## Page 4 · Table block · SPY · one model × four regimes

Same contracts and LSM settings for every model. Bold green row = lowest RMSE%. 1.5-year lookback, rolling=monthly, n\_paths=10000, seed=42.

**Table SPY-1 SPY · 2008-2009 Crisis**  
**2008-08-01 → 2009-07-31 · n = 50 contracts**

| Model | RMSE% | MAE    | Bias \$ | Bias%  | Early-ex. frac. | Mark |
|-------|-------|--------|---------|--------|-----------------|------|
| GARCH | 43.04 | 1.3272 | +1.2328 | +33.41 | 23.5%           | 1    |

**Table SPY-2 SPY · 2013-2014 Normal**  
**2014-01-01 → 2014-12-31 · n = 50 contracts**

| Model | RMSE% | MAE    | Bias \$ | Bias%  | Early-ex. frac. | Mark |
|-------|-------|--------|---------|--------|-----------------|------|
| GARCH | 36.35 | 0.6565 | +0.4203 | +22.86 | 24.4%           | 1    |

**Table SPY-3 SPY · 2018-2019 Late-cycle**  
**2018-10-01 → 2019-09-30 · n = 50 contracts**

| Model | RMSE% | MAE    | Bias \$ | Bias%  | Early-ex. frac. | Mark |
|-------|-------|--------|---------|--------|-----------------|------|
| GARCH | 65.98 | 1.9872 | +1.5638 | +41.26 | 22.2%           | 1    |

**Table SPY-4 SPY · 2019-2020 COVID**  
**2019-09-01 → 2020-08-31 · n = 50 contracts**

| Model | RMSE% | MAE    | Bias \$ | Bias%  | Early-ex. frac. | Mark |
|-------|-------|--------|---------|--------|-----------------|------|
| GARCH | 53.62 | 2.8957 | +2.0456 | +35.24 | 29.1%           | 1    |

## Page 5 · Table block · AAPL · one model × four regimes

Same contracts and LSM settings for every model. Bold green row = lowest RMSE%. 1.5-year lookback, rolling=monthly, n\_paths=10000, seed=42.

**Table AAPL-1 AAPL · 2008-2009 Crisis**  
2008-08-01 → 2009-07-31 · n = 50 contracts

| Model | RMSE% | MAE    | Bias \$ | Bias%  | Early-ex. frac. | Mark |
|-------|-------|--------|---------|--------|-----------------|------|
| GARCH | 34.36 | 1.7480 | +1.5066 | +25.46 | 22.3%           | 1    |

**Table AAPL-2 AAPL · 2013-2014 Normal**  
2014-01-01 → 2014-12-31 · n = 50 contracts

| Model | RMSE% | MAE    | Bias \$ | Bias%  | Early-ex. frac. | Mark |
|-------|-------|--------|---------|--------|-----------------|------|
| GARCH | 40.08 | 2.7582 | +2.5039 | +23.86 | 24.6%           | 1    |

**Table AAPL-3 AAPL · 2018-2019 Late-cycle**  
2018-10-01 → 2019-09-30 · n = 51 contracts

| Model | RMSE% | MAE    | Bias \$ | Bias%  | Early-ex. frac. | Mark |
|-------|-------|--------|---------|--------|-----------------|------|
| GARCH | 40.82 | 1.7245 | +1.1159 | +21.75 | 22.4%           | 1    |

**Table AAPL-4 AAPL · 2019-2020 COVID**  
2019-09-01 → 2020-08-31 · n = 50 contracts

| Model | RMSE% | MAE    | Bias \$ | Bias%  | Early-ex. frac. | Mark |
|-------|-------|--------|---------|--------|-----------------|------|
| GARCH | 45.53 | 3.4285 | +2.6090 | +29.35 | 23.4%           | 1    |

## Page 6 · Table block · MSFT · one model × four regimes

Same contracts and LSM settings for every model. Bold green row = lowest RMSE%. 1.5-year lookback, rolling=monthly, n\_paths=10000, seed=42.

**Table MSFT-1 MSFT · 2008-2009 Crisis**  
2008-08-01 → 2009-07-31 · n = 50 contracts

| Model | RMSE% | MAE    | Bias \$ | Bias%  | Early-ex. frac. | Mark |
|-------|-------|--------|---------|--------|-----------------|------|
| GARCH | 50.93 | 0.4130 | +0.3505 | +35.87 | 28.8%           | 1    |

**Table MSFT-2 MSFT · 2013-2014 Normal**  
2014-01-01 → 2014-12-31 · n = 50 contracts

| Model | RMSE% | MAE    | Bias \$ | Bias%  | Early-ex. frac. | Mark |
|-------|-------|--------|---------|--------|-----------------|------|
| GARCH | 85.94 | 0.6129 | +0.6119 | +73.69 | 25.1%           | 1    |

**Table MSFT-3 MSFT · 2018-2019 Late-cycle**  
2018-10-01 → 2019-09-30 · n = 50 contracts

| Model | RMSE% | MAE    | Bias \$ | Bias%  | Early-ex. frac. | Mark |
|-------|-------|--------|---------|--------|-----------------|------|
| GARCH | 37.25 | 0.9394 | +0.7118 | +24.44 | 21.5%           | 1    |

**Table MSFT-4 MSFT · 2019-2020 COVID**  
2019-09-01 → 2020-08-31 · n = 50 contracts

| Model | RMSE% | MAE    | Bias \$ | Bias%  | Early-ex. frac. | Mark |
|-------|-------|--------|---------|--------|-----------------|------|
| GARCH | 49.20 | 2.0213 | +1.2390 | +25.36 | 25.8%           | 1    |

## Page 7 · Table block · AMZN · one model × four regimes

Same contracts and LSM settings for every model. Bold green row = lowest RMSE%. 1.5-year lookback, rolling=monthly, n\_paths=10000, seed=42.

**Table AMZN-1 AMZN · 2008-2009 Crisis**  
2008-08-01 → 2009-07-31 · n = 50 contracts

| Model | RMSE% | MAE    | Bias \$ | Bias%  | Early-ex. frac. | Mark |
|-------|-------|--------|---------|--------|-----------------|------|
| GARCH | 27.67 | 1.0224 | +0.3210 | +10.57 | 20.9%           | 1    |

**Table AMZN-2 AMZN · 2013-2014 Normal**  
2014-01-01 → 2014-12-31 · n = 50 contracts

| Model | RMSE% | MAE    | Bias \$ | Bias%  | Early-ex. frac. | Mark |
|-------|-------|--------|---------|--------|-----------------|------|
| GARCH | 34.57 | 2.9170 | +1.6120 | +19.87 | 23.0%           | 1    |

**Table AMZN-3 AMZN · 2018-2019 Late-cycle**  
2018-10-01 → 2019-09-30 · n = 51 contracts

| Model | RMSE% | MAE     | Bias \$ | Bias%  | Early-ex. frac. | Mark |
|-------|-------|---------|---------|--------|-----------------|------|
| GARCH | 32.44 | 16.1268 | +9.7576 | +19.15 | 21.2%           | 1    |

**Table AMZN-4 AMZN · 2019-2020 COVID**  
2019-09-01 → 2020-08-31 · n = 50 contracts

| Model | RMSE% | MAE     | Bias \$ | Bias%  | Early-ex. frac. | Mark |
|-------|-------|---------|---------|--------|-----------------|------|
| GARCH | 30.12 | 18.8538 | +3.8848 | +15.17 | 22.1%           | 1    |

## Page 7 · Summary tables · RMSE% across companies and regimes

**Table S1 · Overall ranking (mean RMSE% over 16 cells)**

| Rank | Model | Mean RMSE% | Median RMSE% | # of 16 cells best |
|------|-------|------------|--------------|--------------------|
| 1    | GARCH | 44.24      | 40.45        | 16                 |

**Table S2 · Wins by company (lowest RMSE% in that company×regime)**

| Model | SPY wins | AAPL wins | MSFT wins | AMZN wins | Total |
|-------|----------|-----------|-----------|-----------|-------|
| GARCH | 4        | 4         | 4         | 4         | 16    |

**Table S3 · Mean RMSE% by regime (equal-weight mean of SPY / AAPL / MSFT / AMZN)**

| Model | 2008-2009<br>Crisis | 2013-2014<br>Normal | 2018-2019<br>Late-cycle | 2019-2020<br>COVID | Mean of 4 |
|-------|---------------------|---------------------|-------------------------|--------------------|-----------|
| GARCH | 39.00               | 49.23               | 44.12                   | 44.61              | 44.24     |

Green row in S1 / S3 = lowest mean RMSE%. Green row in S2 = most company×regime wins. A model can win the most cells without having the lowest average error (and vice versa).



Table S4.1 · SPY

| Model | Mean RMSE% | Median | # regimes best |
|-------|------------|--------|----------------|
| GARCH | 49.75      | 48.33  | 4              |

Table S4.2 · AAPL

| Model | Mean RMSE% | Median | # regimes best |
|-------|------------|--------|----------------|
| GARCH | 40.20      | 40.45  | 4              |

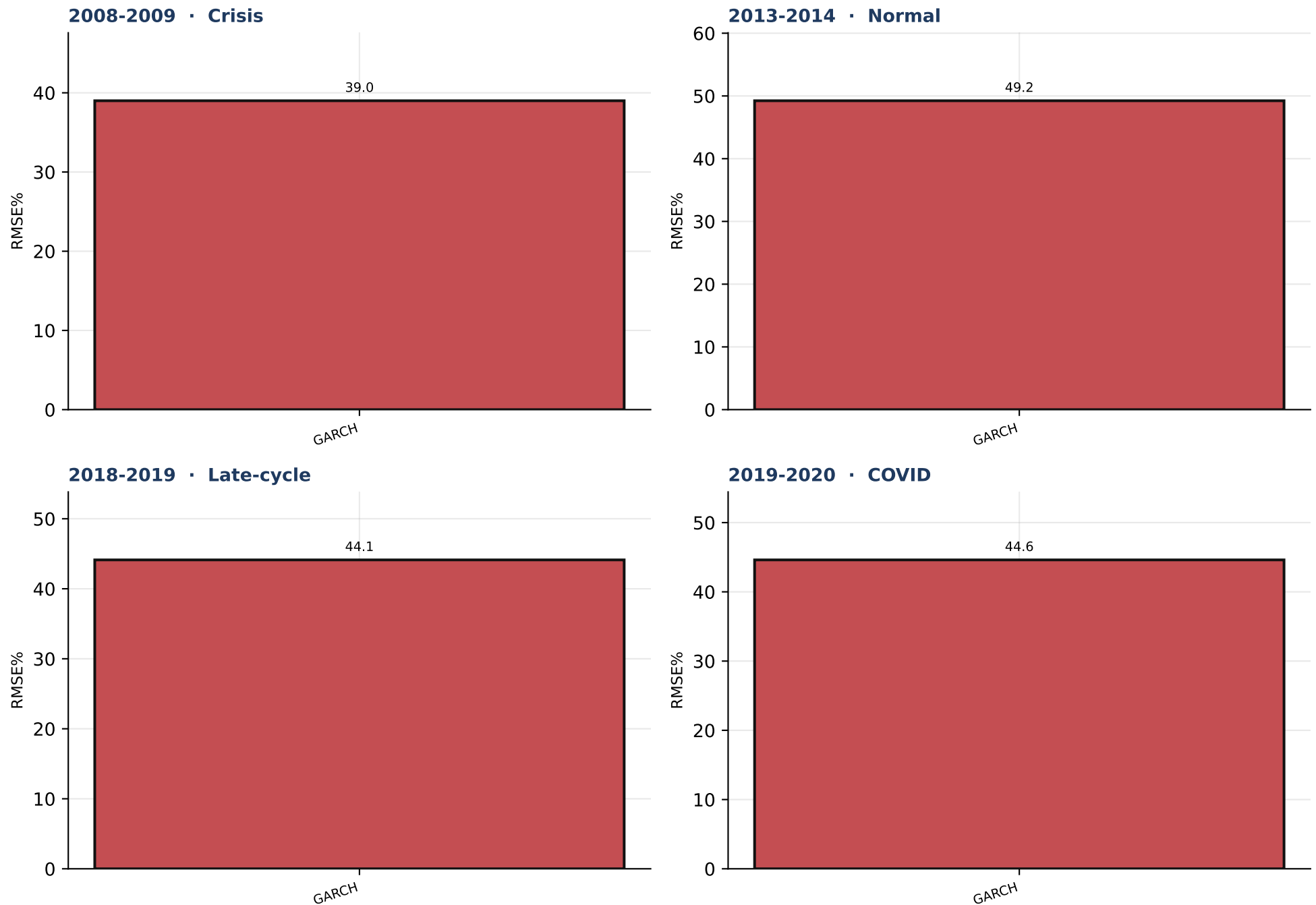
Table S4.3 · MSFT

| Model | Mean RMSE% | Median | # regimes best |
|-------|------------|--------|----------------|
| GARCH | 55.83      | 50.06  | 4              |

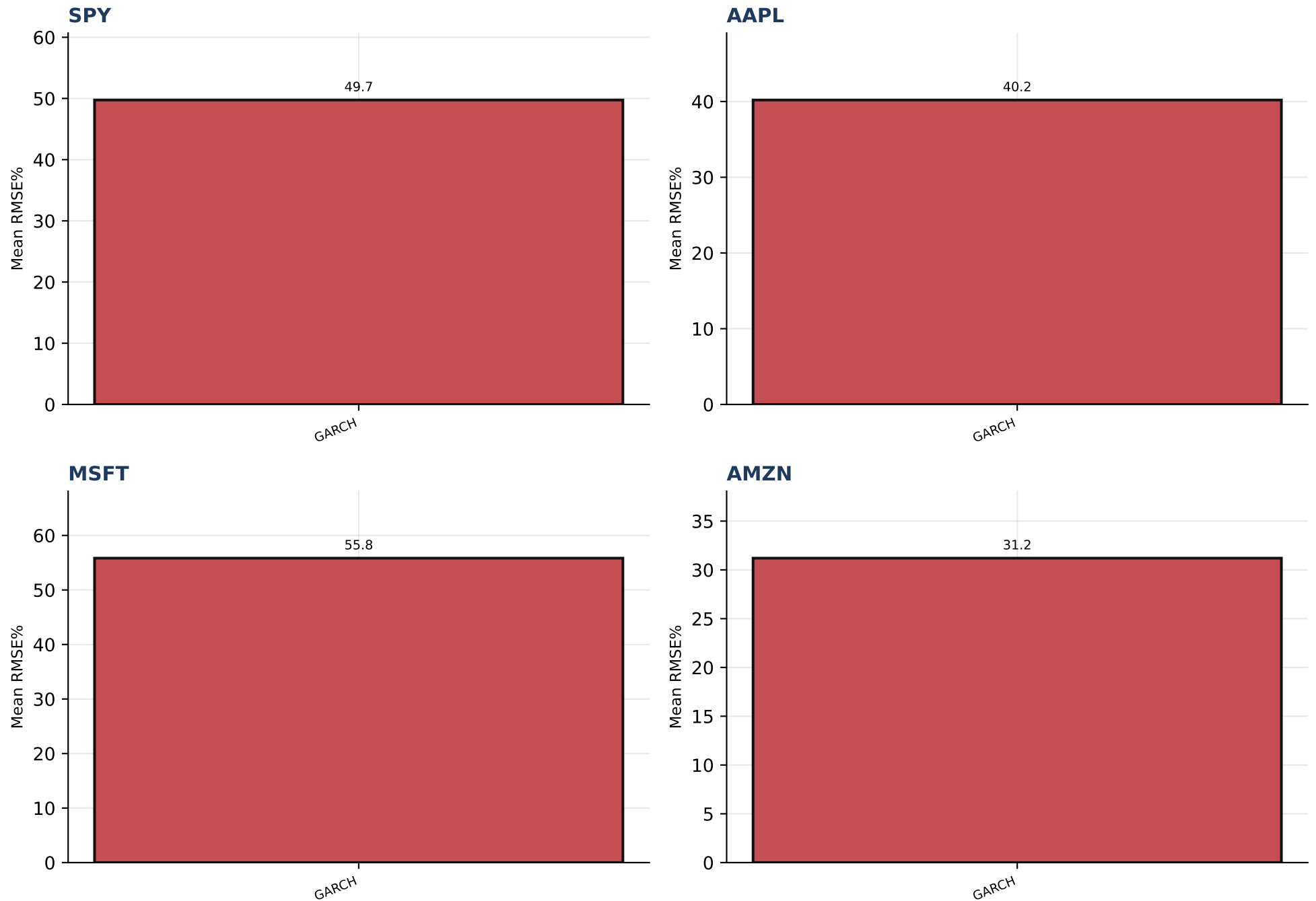
Table S4.4 · AMZN

| Model | Mean RMSE% | Median | # regimes best |
|-------|------------|--------|----------------|
| GARCH | 31.20      | 31.28  | 4              |

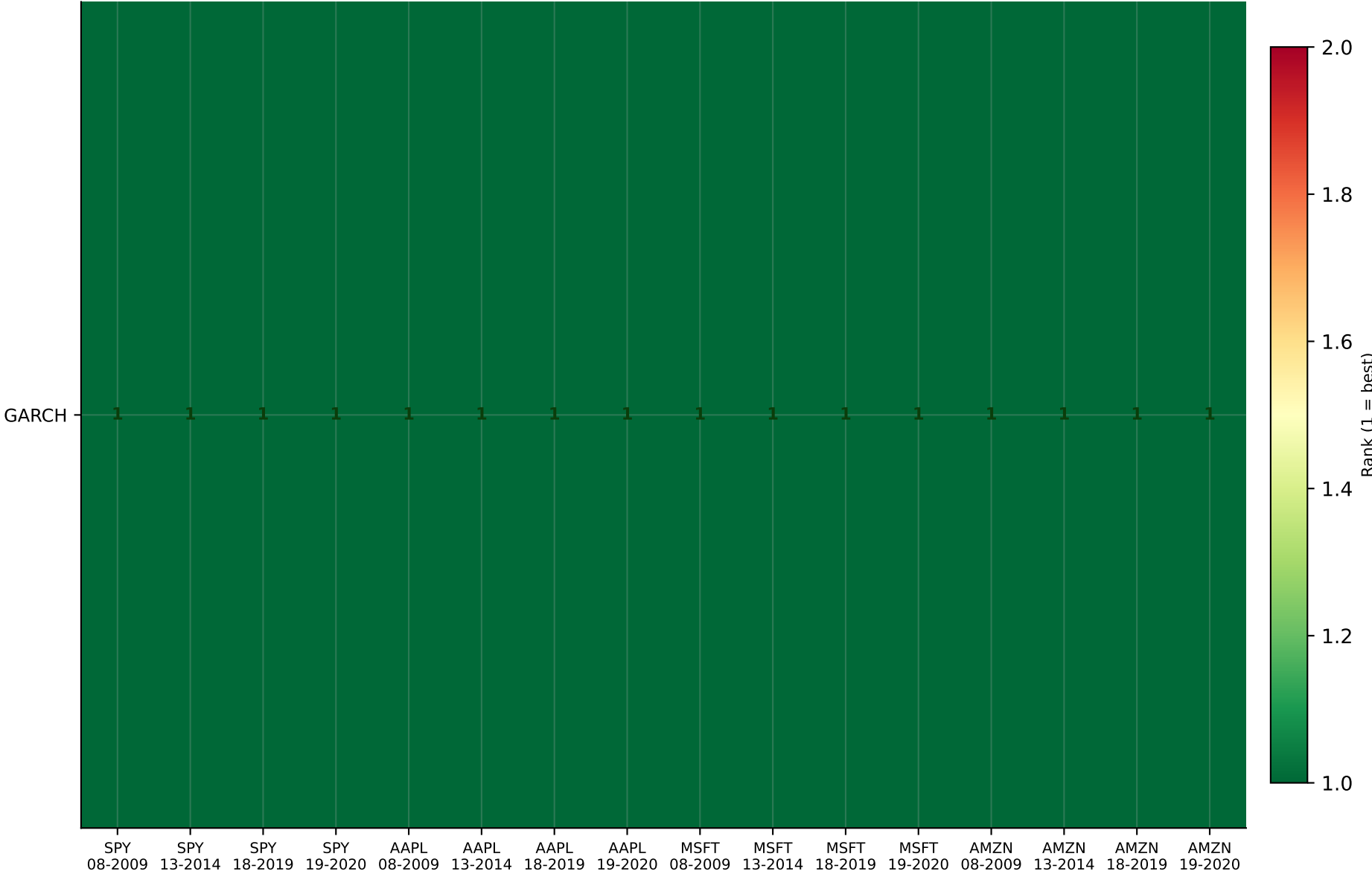
Each panel is one company. Green row = lowest mean RMSE% across the four V3 regimes for that name.



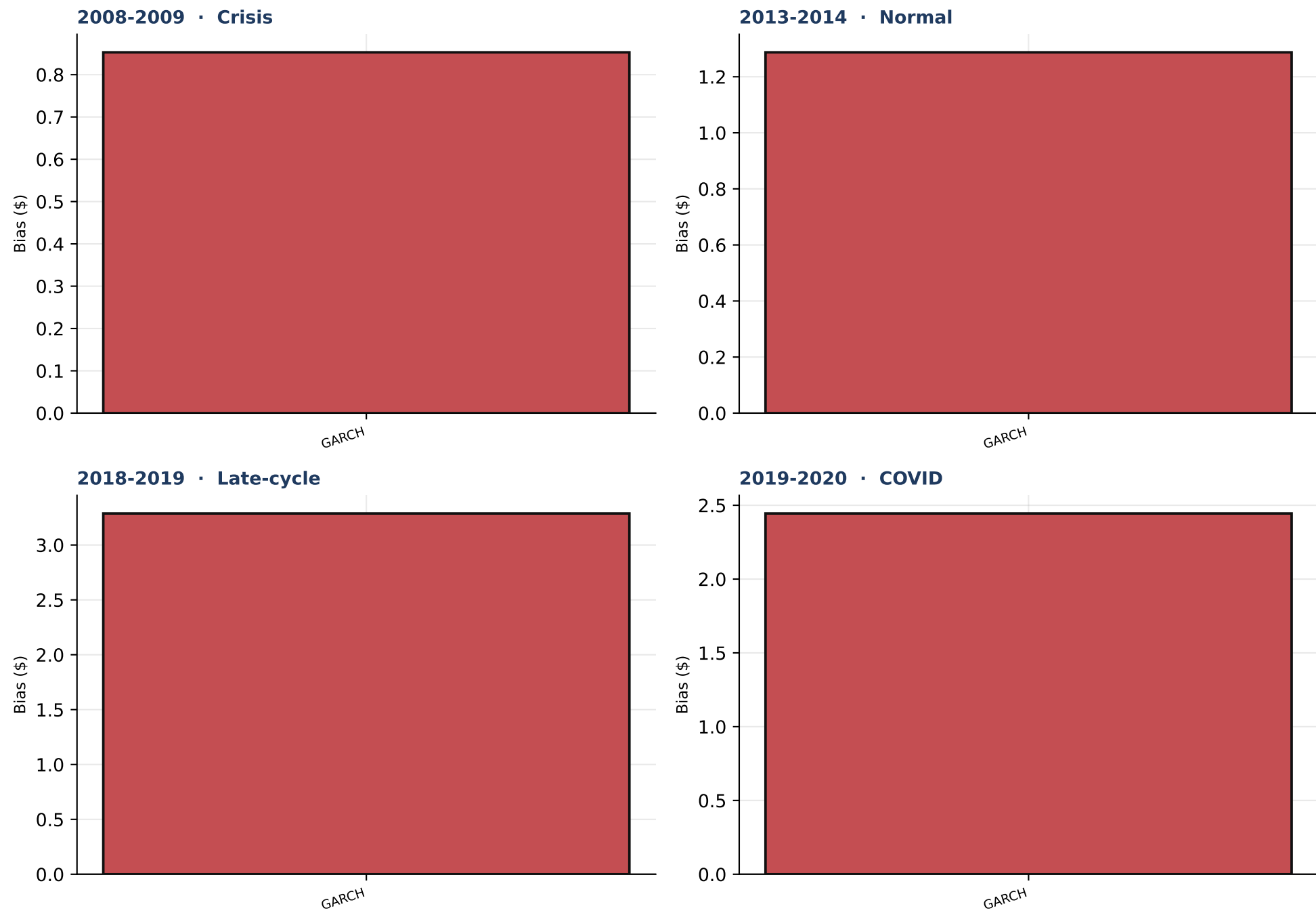
Outlined bar = lowest mean RMSE% in that regime. Equal-weight mean of the four companies.



Outlined bar = lowest mean RMSE% for that company.



Green / 1 = lowest RMSE% in that company×regime cell. Rankings can and do change across names and volatility states.



Outlined bar = closest to zero bias (not the RMSE ranking). Positive = model expensive vs listed quote.

### Which models perform best, and does the ranking change?

- On mean percentage RMSE across the 16 company×regime cells, GARCH is the best overall model (mean RMSE% = 44.24; 16 of 16 cells).
- The ranking is stable: GARCH is best in every company and every regime.
- SPY: lowest mean RMSE% is GARCH (49.75); wins 4 of 4 regimes.
- AAPL: lowest mean RMSE% is GARCH (40.20); wins 4 of 4 regimes.
- MSFT: lowest mean RMSE% is GARCH (55.83); wins 4 of 4 regimes.
- AMZN: lowest mean RMSE% is GARCH (31.20); wins 4 of 4 regimes.
- 2008-2009 (Crisis): best mean RMSE% is GARCH (39.00).
- 2013-2014 (Normal): best mean RMSE% is GARCH (49.23).
- 2018-2019 (Late-cycle): best mean RMSE% is GARCH (44.12).
- 2019-2020 (COVID): best mean RMSE% is GARCH (44.61).
- For the overall winner (GARCH), crisis RMSE% is 39.00 vs 49.23 in the 2014 normal year — absolute pricing error is regime-dependent even when the ranking is not.
- These conclusions are conditional on the V3 filters, 1.5-year monthly calibration, Monday ATM sample, LSM with 10,000 paths, and American-call quotes only. They are not a statement about European options or other tenors.

### How to read the 12 detailed tables

- Each of Tables SPY-1...MSFT-4 uses exactly the same option contracts for all one model in this report. Differences are the dynamics, not the sample.
- BEST is always lowest RMSE%. MAE, Bias \$, Bias%, and early-exercise are reported, not used for the ranking.
- A model that is BEST on RMSE% can still be biased (systematically expensive or cheap).
- Early-exercise fractions are a model implication, not an accuracy score; American calls on these tenors often have modest early-exercise value.